

HONOURS ANALYSIS 1: REVIEW SESSION 2

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1. ABOUT QUANTIFIERS

Let's take a moment to recall the Archimedean property: for all $x \in \mathbb{R}$, there exists $n \in \mathbb{N}$ such that $x \leq n$. Or, in symbolic terms,

$$(\forall x \in \mathbb{R})(\exists n \in \mathbb{N})(x \leq n). \quad (1)$$

This is **not** equivalent to the statement

$$(\exists n \in \mathbb{N})(\forall x \in \mathbb{R})(x \leq n). \quad (2)$$

Indeed, written in words, (2) states that there exists a natural number $n \in \mathbb{N}$ such that, for all $x \in \mathbb{R}$, there holds $x \leq n$. Clearly, one cannot have $x \leq n$ for **all** $x \in \mathbb{R}$ as taking $x = n + 1$ would give $n + 1 \leq n$, a contradiction. So, not only are (1) and (2) not equivalent, but (2) is **false**! The issue is that in (1) the natural number n is allowed to depend on x , whereas in (2) n appears *before* the “ $\forall x \in \mathbb{R}$ ” statement. Thus, in (2), the natural number n does *not* depend on x . Put otherwise, (2) says that the same natural number n can work for *all* $x \in \mathbb{R}$.

We will now go over some of the ungraded (but relevant) problems from assignments 5-8. Please note that several of these problems were already discussed during the tutorials and will not be reiterated here since the notes are already available on MyCourses.

2. PROBLEMS FROM ASSIGNMENT 5

Problem 2.1. *Prove that*

- (1) $\overset{\circ}{\mathbb{Q}} = \emptyset$ and $\text{int}(\mathbb{R} \setminus \mathbb{Q}) = \emptyset$;
- (2) $\overline{\mathbb{Q}} = \mathbb{R}$ and $\overline{\mathbb{R} \setminus \mathbb{Q}} = \mathbb{R}$.

Solution.

- (1) We only show that $\overset{\circ}{\mathbb{Q}} = \emptyset$ since an identical argument also implies that $\text{int}(\mathbb{R} \setminus \mathbb{Q}) = \emptyset$. Now, let's assume for the sake of a contradiction that $\overset{\circ}{\mathbb{Q}} \neq \emptyset$. Then, we can pick $x \in \overset{\circ}{\mathbb{Q}}$. Since the interior of any set is always

open, there exists $\varepsilon > 0$ such that $V_\varepsilon(x) = (x - \varepsilon, x + \varepsilon) \subseteq \mathring{\mathbb{Q}} \subseteq \mathbb{Q}$. However, because $\mathbb{R} \setminus \mathbb{Q}$ is dense in $\mathbb{R}$, there exists $\xi \in \mathbb{R} \setminus \mathbb{Q}$ such that

$$x - \varepsilon < \xi < x + \varepsilon.$$

This implies that $\xi \in V_\varepsilon(x) \subseteq \mathbb{Q}$ which contradicts the irrationality of ξ .

- (2) More generally, it is enough to prove that $\overline{A} = \mathbb{R}$ for any *dense* subset A of $\mathbb{R}$. Letting $B := \overline{A}$, the claim amounts to proving that $B = \mathbb{R}$. Arguing by contradiction, we assume that $B \subsetneq \mathbb{R}$. Thus, there exists a point $x \in B^c$. Since B is closed, B^c must be open. Therefore, $V_\varepsilon(x) \subseteq B^c$ for some $\varepsilon > 0$. Now, since A is dense in $\mathbb{R}$, there exists $a \in A$ satisfying

$$x - \varepsilon < a < x + \varepsilon.$$

Thus, $a \in V_\varepsilon(x) \subseteq B^c$. On the other hand, $a \in A \subseteq \overline{A} = B$. This implies that $B \cap B^c \neq \emptyset$ which is a contradiction.

□

Problem 2.2. Let $S_1, \dots, S_n$ be subsets of $\mathbb{R}$ for some $n \in \mathbb{N}$. Prove that

$$\overline{S_1 \cup \dots \cup S_n} = \overline{S_1} \cup \dots \cup \overline{S_n}.$$

Proof. By definition of the closure, for each index $j = 1, \dots, n$, $\overline{S_j}$ is closed. Since finite unions of closed sets are closed, the union

$$\overline{S_1} \cup \dots \cup \overline{S_n}$$

is a closed set containing $S_1 \cup \dots \cup S_n$. By definition of the closure,

$$\overline{S_1 \cup \dots \cup S_n} = \bigcap_{\substack{F \text{ closed} \\ F \supseteq S_1 \cup \dots \cup S_n}} F \subseteq \overline{S_1} \cup \dots \cup \overline{S_n}.$$

This proves the “ $\subseteq$ ” inclusion. Conversely, fix $j = 1, \dots, n$ and note that

$$S_j \subseteq S_1 \cup \dots \cup S_n \subseteq \overline{S_1 \cup \dots \cup S_n}$$

with $\overline{S_1 \cup \dots \cup S_n}$ being closed. Thus, $\overline{S_j} \subseteq \overline{S_1 \cup \dots \cup S_n}$ for all $j = 1, \dots, n$. Consequently,

$$\overline{S_1} \cup \dots \cup \overline{S_n} \subseteq \overline{S_1 \cup \dots \cup S_n}$$

which completes the proof. □

Remark 1. It is useful to think of $\overline{A}$ as the *smallest* closed set containing A . In particular, if $F \supseteq A$ is any closed set, then we must have $A \subseteq \overline{A} \subseteq F$. This property will be frequently used when discussing the closure.

Problem 2.3. Let $\{X_\alpha\}_{\alpha \in I}$ be an indexed family of subsets of $\mathbb{R}$. Prove that

$$\overline{\bigcap_{\alpha \in I} X_\alpha} \subseteq \bigcap_{\alpha \in I} \overline{X_\alpha}.$$

Proof. Since $X_\alpha \subseteq \overline{X_\alpha}$ for all $\alpha \in I$, it is easily seen that $\bigcap_{\alpha \in I} X_\alpha \subseteq \bigcap_{\alpha \in I} \overline{X_\alpha}$.¹ Now, as arbitrary intersections of closed sets are closed, we see that $\bigcap_{\alpha \in I} \overline{X_\alpha}$ is a closed set containing $\bigcap_{\alpha \in I} X_\alpha$. By definition of the closure, we infer that

$$\overline{\bigcap_{\alpha \in I} X_\alpha} \subseteq \bigcap_{\alpha \in I} \overline{X_\alpha},$$

as was required. □

Remark 2. In the above problem it is not certain that equality holds, even when the intersections are finite. Consider for instance the sets $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$. We see that the closure of their intersection does not correspond to the intersection of the closures. Indeed, $\overline{\mathbb{Q} \cap [\mathbb{R} \setminus \mathbb{Q}]} = \overline{\emptyset} = \emptyset$ while $\overline{\mathbb{Q}} \cap \overline{\mathbb{R} \setminus \mathbb{Q}} = \mathbb{R} \cap \mathbb{R} = \mathbb{R}$.

Problem 2.4. Prove the following:

- (1) if $U \subseteq \mathbb{R}$ is open, then ∂U has empty interior;
- (2) if $A \subseteq \mathbb{R}$ is closed, then ∂A has empty interior;
- (3) give an example of a set $S \subseteq \mathbb{R}$ such that ∂S has non-empty interior.

Proof.

- (1) If $\text{int}(\partial U)$ contains even a single point x , then we can find $\varepsilon > 0$ such that $V_\varepsilon(x) \subseteq \text{int}(\partial U) \subseteq \partial U$. Now, using that U is open, a previous assignment tells us that $\partial U \cap U = \emptyset$. Equivalently, $\partial U \subseteq U^c$ whence $V_\varepsilon(x) \subseteq \partial U \subseteq U^c$. But, $x \in \text{int}(\partial U) \subseteq \partial U$ implies that

$$V_\varepsilon(x) \cap U \neq \emptyset \quad \text{and} \quad V_\varepsilon(x) \cap U^c \neq \emptyset.$$

In particular, $V_\varepsilon(x) \cap U \neq \emptyset$ which contradicts the fact that $V_\varepsilon(x) \subseteq U^c$.

- (2) If A is closed then A^c is open whence $\partial(A^c)$ has empty interior by the first part. By an earlier assignment, we know that $\partial A = \partial(A^c)$. Especially, ∂A has empty interior.
- (3) We have seen in the lectures that $\partial \mathbb{Q} = \mathbb{R}$. Thus, $\partial \mathbb{Q}$ has non-empty interior. □

¹Indeed, if $x \in \bigcap_{\alpha \in I} X_\alpha$ then $x \in X_\alpha$ for all $\alpha \in I$. Since $X_\alpha \subseteq \overline{X_\alpha}$, this implies that $x \in \bigcap_{\alpha \in I} \overline{X_\alpha}$.

3. ASSIGNMENT 6

Problem 3.1. *Using the definition of the limit, show that*

$$\lim \left(\sqrt{n^2 + 1} - n \right) = 0.$$

Proof. Let $\varepsilon > 0$ be given, we seek $N \in \mathbb{N}$ such that

$$\left| \sqrt{n^2 + 1} - n \right| < \varepsilon, \quad \forall n \geq N.$$

Now, for any $n \geq 1$, multiplying by the conjugate yields

$$\begin{aligned} \left| \sqrt{n^2 + 1} - n \right| &= \sqrt{n^2 + 1} - \sqrt{n^2} = \frac{(\sqrt{n^2 + 1} - \sqrt{n^2})(\sqrt{n^2 + 1} + \sqrt{n^2})}{\sqrt{n^2 + 1} + \sqrt{n^2}} \\ &= \frac{1}{\sqrt{n^2 + 1} + \sqrt{n^2}} \\ &\leq \frac{1}{\sqrt{n^2}} = \frac{1}{n}. \end{aligned}$$

By the Archimedean property, there exists $N \in \mathbb{N}$ such that $N > \frac{1}{\varepsilon}$. Then, $\frac{1}{N} < \varepsilon$. By our calculations above, if $n \geq N$ one has

$$\left| \sqrt{n^2 + 1} - n \right| \leq \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

□

Problem 3.2. *Using the definition of the limit, show that*

$$\lim \left(\frac{\sqrt{n} + (-1)^n}{\sqrt{n} + 1} \right) = 1.$$

Proof. Let $\varepsilon > 0$ be given. For any $n \in \mathbb{N}$, one has by the triangle inequality that

$$\begin{aligned} \left| \frac{\sqrt{n} + (-1)^n}{\sqrt{n} + 1} - 1 \right| &= \left| \frac{\sqrt{n} + (-1)^n - \sqrt{n} - 1}{\sqrt{n} + 1} \right| \\ &= \frac{|(-1)^n - 1|}{\sqrt{n} + 1} \\ &\leq \frac{|(-1)^n| + |-1|}{\sqrt{n} + 1} \\ &= \frac{2}{\sqrt{n} + 1} \\ &\leq \frac{2}{\sqrt{n}}. \end{aligned}$$

Again using the Archimedean property, there exists $N \in \mathbb{N}$ such that $N > \frac{4}{\varepsilon^2}$. Clearly, $\sqrt{N} > \frac{2}{\varepsilon}$ whence, for all $n \geq N$,

$$\left| \frac{\sqrt{n} + (-1)^n}{\sqrt{n} + 1} - 1 \right| \leq \frac{2}{\sqrt{n}} \leq \frac{2}{\sqrt{N}} < \varepsilon.$$

□

4. ASSIGNMENT 7

Problem 4.1. Let (b_n) be a bounded sequence and assume that $\lim a_n = 0$. Prove that $\lim a_n b_n = 0$.

Proof. Since (b_n) is bounded, there exists $M > 0$ such that $|b_n| \leq M$ for all $n \in \mathbb{N}$. Fix $\varepsilon > 0$. Using that $a_n \rightarrow 0$, there exists $N \in \mathbb{N}$ such that

$$|a_n| < \frac{\varepsilon}{M}$$

for all $n \geq N$. Then, for all $n \geq N$,

$$|a_n b_n - 0| = |a_n| |b_n| \leq M |a_n| < M \cdot \frac{\varepsilon}{M} = \varepsilon.$$

This concludes the proof. □

Problem 4.2. Using the Squeeze theorem, determine the following limits:

- (1) $\lim \left(n^{1/n^2} \right);$
- (2) $\lim \left((n!)^{1/n^2} \right).$

Solution. From Assignment 7, we know that $\lim \sqrt[n]{n} = 1$.

- (1) For all $n \in \mathbb{N}$ we have $\frac{1}{n^2} \leq \frac{1}{n}$. Hence,

$$1 \leq n^{1/n^2} \leq n^{1/n}, \quad \forall n \in \mathbb{N}.$$

By the Squeeze theorem, we infer that $\lim n^{1/n^2} = 1$.

- (2) Here we argue similarly. Note that, for all $n \in \mathbb{N}$,

$$n! = n(n-1)(n-2) \cdots 2 \cdot 1 \leq \underbrace{n \cdot n \cdots n}_{n\text{-times}} = n^n.$$

Therefore, for each $n \geq 1$ there holds

$$1 \leq (n!)^{1/n^2} \leq (n^n)^{1/n^2} = n^{1/n}.$$

Again citing the squeeze theorem, we see that the limit in question is equal to 1. □

Problem 4.3. Let $x_1 > 0$ and define for all $n \in \mathbb{N}$ the quantity

$$x_{n+1} := x_n + \frac{1}{x_n}.$$

Prove that the sequence (x_n) is unbounded.

Proof. Owing to the fact that $x_1 > 0$, it is clear that $x_n > 0$ for all $n \geq 1$. Moreover, (x_n) is an increasing sequence. Indeed, using that $x_n > 0$, we obtain

$$x_{n+1} = x_n + \underbrace{\frac{1}{x_n}}_{>0} > x_n.$$

If (x_n) were bounded, then it would also be convergent (by the monotone convergence theorem) with limit $x \geq 0$. Now, the equation

$$x_{n+1} = x_n + \frac{1}{x_n}$$

implies that $x_n x_{n+1} = x_n^2 + 1$, for all $n \in \mathbb{N}$. If $x_n \rightarrow x$, then the limit laws tell us that

$$x^2 = \lim x_n x_{n+1} = \lim (x_n^2 + 1) = \lim x_n^2 + 1 = x^2 + 1.$$

As this yields $0 = 1$, we have a contradiction. Hence, (x_n) cannot be bounded. $\square$

5. ASSIGNMENT 8 (AND ADDITIONAL PROBLEMS)

Problem 5.1. Let (x_n) be a bounded sequence and let A be the set of all accumulation points of (x_n) . Then, A is closed.

Proof. We must show that A^c is open. To this end, fix a point $x \in A^c$. Since x is not an accumulation point of (x_n) , there exists $\varepsilon > 0$ such that $V_\varepsilon(x)$ contains only finitely many x_n . If we can show that $V_\varepsilon(x) \subseteq A^c$ then we will have shown that A^c is open. Now, for any $y \in V_\varepsilon(x)$ there exists $\delta > 0$ such that $V_\delta(y) \subseteq V_\varepsilon(x)$. In particular, $V_\delta(y)$ contains only finitely many x_n . This means that y is *not* an accumulation point of (x_n) . Since $y \in V_\varepsilon(x)$ was arbitrary, we infer that $V_\varepsilon(x) \subseteq A^c$. $\square$

Problem 5.2. There are no finite dense subsets of $\mathbb{R}$. Moreover, if A is dense in $\mathbb{R}$, so is $A \setminus \{x\}$ for any $x \in \mathbb{R}$.

Proof. Recall that finite sets are closed. Moreover, we proved in Problem 2.1 that $\overline{A} = \mathbb{R}$ whenever A is dense in $\mathbb{R}$. Thus, if A is a finite dense subset of $\mathbb{R}$, we obtain $A = \overline{A} = \mathbb{R}$ whence A is infinite. This contradiction shows that $\mathbb{R}$ has no finite dense subsets. Next, we fix $x \in \mathbb{R}$ and show that $A \setminus \{x\}$ is dense in $\mathbb{R}$. For this, it is enough to show that $U \cap (A \setminus \{x\}) \neq \emptyset$ for all non-empty open sets $U \subseteq \mathbb{R}$. However,

if U is open, then so is $U \setminus \{x\} = U \cap \{x\}^c$.² Since A is dense in $\mathbb{R}$, there exists a point

$$a \in A \cap (U \setminus \{x\}).$$

In particular, $a \in U \setminus \{x\}$ so that $a \in U$ but $a \neq x$. Therefore, $a \in (A \setminus \{x\})$ and $a \in U$. This means that $(A \setminus \{x\}) \cap U \neq \emptyset$ and the claim is thus proven. $\square$

Remark 3. Let $A \subseteq \mathbb{R}$ be dense. By induction, the result above says that $A \setminus S$ is still dense in $\mathbb{R}$ for any finite set $S \subset \mathbb{R}$.

Problem 5.3. Let (x_n) be any enumeration of $\mathbb{Q}$ (possibly having repetitions). Denote by A the set of all accumulation points of (x_n) . Prove that $A = \mathbb{R}$.

Proof. Clearly, $A \subseteq \mathbb{R}$. Conversely, fix a point $x \in \mathbb{R}$. We claim that $x \in A$. Since A consists of all accumulation points of (x_n) , we must find a subsequence (x_{n_k}) of (x_n) converging to x . Since $\mathbb{Q}$ is dense in $\mathbb{R}$, there exists a term x_{n_1} in our sequence (x_n) such that $x_{n_1} \in (x - 1, x + 1) = V_1(x)$. Having chosen $x_{n_k} \in V_{1/k}(x)$, note that

$$\mathbb{Q} \setminus \bigcup_{j=1}^{n_k} \{x_j\}$$

is still dense in $\mathbb{R}$ by our remark above. Hence,

$$\left(\mathbb{Q} \setminus \bigcup_{j=1}^{n_k} \{x_j\} \right) \cap V_{\frac{1}{k+1}}(x) \neq \emptyset.$$

In particular, there exists a rational number $r \in V_{\frac{1}{k+1}}(x)$ such that $r \neq x_j$ for all $j = 1, \dots, n_k$. Since (x_n) is an enumeration of $\mathbb{Q}$, there exists some index $m \in \mathbb{N}$ with $x_m = r$. However, since $r \neq x_j$ for all $j = 1, \dots, n_k$, we have $m > n_k$. We then define $x_{n_{k+1}} = x_m = r$.

This procedure gives us a subsequence (x_{n_k}) of (x_n) with $x_{n_k} \in V_{1/k}(x)$ for all $k \geq 1$. Put otherwise, we have

$$|x_{n_k} - x| < \frac{1}{k}, \quad \forall k \in \mathbb{N}.$$

The Squeeze theorem then implies that $x_{n_k} \rightarrow x$ as $n_k \rightarrow \infty$. We have therefore found a subsequence (x_{n_k}) of (x_n) that converges to x , and so $x \in A$. $\square$

Problem 5.4. Consider the sequence

$$x_n = \left(1 + \frac{(-1)^n}{n} \right)^n.$$

Show that (x_n) is bounded, determine all the accumulation points of (x_n) , and compute $\limsup x_n$ and $\liminf x_n$.

²Note that $U \setminus \{x\}$ cannot be empty since non-empty open sets are uncountable.

Solution. We saw in the tutorials that

$$\left(1 - \frac{1}{n}\right)^n \xrightarrow{n \rightarrow \infty} e^{-1}.$$

Clearly,

$$x_{2n+1} = \left(1 - \frac{1}{2n+1}\right)^{2n+1}$$

is a subsequence of $\left(1 - \frac{1}{n}\right)^n$ and so (x_{2n+1}) must converge to e^{-1} . Similarly, (x_{2n}) is a subsequence of $\left(1 + \frac{1}{n}\right)^n$ and so $x_{2n} \rightarrow e$ as $n \rightarrow \infty$. Especially, both (x_{2n}) and (x_{2n+1}) are bounded. This means there exists $M > 0$ such that

$$|x_{2n}| \leq M \quad \text{and} \quad |x_{2n+1}| \leq M$$

for all $n \in \mathbb{N}$. Since every $n \in \mathbb{N}$ is either of the form $2k$ or $2k+1$ for some $k \in \mathbb{N}$, this implies that $|x_n| \leq M$ for all $n \geq 1$. That is, (x_n) is bounded. Now, as (x_{2n}) and (x_{2n+1}) are subsequences of (x_n) , it is immediate that

$$\text{Accumulation}(x_n) \supseteq \{e^{-1}, e\}.$$

We now claim that e^{-1} and e are the only accumulation points of (x_n) . To this end, let x be an accumulation point of (x_n) and let (x_{n_k}) be a subsequence of (x_n) converging to x . Since (x_{n_k}) must share a common subsequence with (x_{2n}) or (x_{2n+1}) , we must have

$$x = \lim x_{2n} \quad \text{or} \quad x = \lim x_{2n+1}.$$

That is, $x = e$ or $x = e^{-1}$. This shows that $\text{Accumulation}(x_n) = \{e, e^{-1}\}$. It follows that

$$\limsup x_n = \sup \{e, e^{-1}\} = e \quad \text{and} \quad \liminf x_n = \inf \{e, e^{-1}\} = e^{-1}.$$

□

Problem 5.5. Let (x_n) be a bounded sequence in $\mathbb{R}$.

- (1) Prove that (x_n) is convergent if and only if it has exactly one accumulation point.
- (2) Prove that (x_n) converges if and only if

$$\limsup x_n = \liminf x_n.$$

Moreover, if (x_n) is convergent, then $\lim x_n = \limsup x_n = \liminf x_n$.

- (3) Give an example of a (possibly unbounded) sequence (x_n) that does not converge but has exactly one accumulation point.

Solution.

- (1) If (x_n) converges to x , then so does every subsequence of (x_n) . This means that x is the only accumulation point of (x_n) . Conversely, assume that (x_n) has only a single accumulation point, and call it x . By way of contradiction, assume that $x_n \not\rightarrow x$. For some $\varepsilon_0 > 0$ and all $N \in \mathbb{N}$, there exists $n \geq N$ such that $|x_n - x| \geq \varepsilon_0$. This guarantees the existence of a subsequence (x_{n_k}) of (x_n) such that

$$|x_{n_k} - x| \geq \varepsilon_0 > 0, \quad \forall k \in \mathbb{N}. \quad (3)$$

Since (x_{n_k}) is a subsequence of a bounded sequence, it is itself bounded. By the Bolzano-Weierstrass theorem, we can find a subsequence $(x_{n_{k_j}})$ of (x_{n_k}) converging to some $x^* \in \mathbb{R}$. However, (3) shows that

$$|x^* - x| = \lim_{j \rightarrow \infty} |x_{n_{k_j}} - x| \geq \varepsilon_0$$

whence $x^* \neq x$. Since $(x_{n_{k_j}})$ is also a subsequence of (x_n) , this means that $x^* \neq x$ is another accumulation point of (x_n) – which is a contradiction.

- (2) If (x_n) converges, then it has exactly one accumulation point by (1). If we denote this accumulation point by x , then it follows from the definition for $\limsup$ and $\liminf$ that

$$\limsup x_n = x = \liminf x_n.$$

Conversely, assume that $\limsup x_n = \liminf x_n$. If x is any accumulation point of (x_n) , we have

$$\liminf x_n \leq x \leq \limsup x_n = \liminf x_n$$

whence $x = \liminf x_n = \limsup x_n$. Thus, (x_n) has only one accumulation point. Once again, it follows from (1) that (x_n) converges.

- (3) Consider the sequence

$$x_n := \begin{cases} n & \text{if } n \text{ is odd,} \\ 1 & \text{if } n \text{ is even.} \end{cases}$$

Then, (x_n) is unbounded and has a subsequence (x_{2n}) converging to 1. Thus, 1 is an accumulation point of (x_n) . Let now (x_{n_k}) be a convergent subsequence of (x_n) (in particular, (x_{n_k}) is bounded). As before, (x_{n_k}) shares a common subsequence with (x_{2n}) or (x_{2n+1}) . However, if (x_{n_k}) were to contain a subsequence of (x_{2n+1}) , then (x_{n_k}) would be unbounded. Thus, (x_{n_k}) shares a common subsequence with (x_{2n}) . Since (x_{n_k}) and (x_{2n}) are convergent, it follows that

$$\lim x_{n_k} = \lim x_{2n} = 1.$$

As (x_{n_k}) was an arbitrary convergent subsequence of (x_n) , we see that 1 is the only accumulation point of (x_n) . □

Problem 5.6. Let (x_n) and (y_n) be two bounded sequences. Prove that

$$\liminf(x_n + y_n) \leq \liminf x_n + \limsup y_n.$$

Proof. We know from class that there exists a subsequence (x_{n_k}) of (x_n) converging to $\liminf x_n$. Now, as (y_n) is bounded, there exists by Bolzano-Weierstrass a subsequence $(y_{n_{k_j}})$ converging to some point $y \in \mathbb{R}$. Clearly, $y \leq \limsup y_n$ and $\lim x_{n_{k_j}} = \lim x_{n_k} = \liminf x_n$. Therefore, by the limit laws,

$$\begin{aligned} \lim(x_{n_{k_j}} + y_{n_{k_j}}) &= \lim x_{n_{k_j}} + \lim y_{n_{k_j}} = \liminf x_n + y \\ &\leq \liminf x_n + \limsup y_n. \end{aligned}$$

Since $(x_{n_{k_j}} + y_{n_{k_j}})$ is a subsequence of $(x_n + y_n)$, it follows that

$$\liminf(x_n + y_n) \leq \liminf x_n + \limsup y_n.$$
□

Problem 5.7. Let (x_n) and (y_n) be bounded sequences such that

$$\liminf x_n = 1 \quad \text{and} \quad \liminf y_n = 1.$$

If $(x_n + y_n)$ is convergent with $\lim(x_n + y_n) = 2$, then $\lim x_n = 1$ and $\lim y_n = 1$.

Proof. Since the problem is symmetric in (x_n) and (y_n) , it suffices to check that $\lim x_n = 1$. Suppose by way of contradiction that $\lim x_n \neq 1$. Then, by Problem 5.4 we must have $\limsup x_n > 1$. Let (x_{n_k}) be a subsequence of (x_n) converging to $x = \limsup x_n > 1$. Since (y_n) is once again bounded, there exists a convergent subsequence $(y_{n_{k_j}})$ of (y_{n_k}) with limit y . Since y is also an accumulation point of (y_n) , we must have

$$y \geq \liminf y_n = 1.$$

Now, as the subsequence of a convergent sequence, we have

$$\lim x_{n_{k_j}} = \lim x_{n_k} = x.$$

Thus,

$$\lim(x_{n_{k_j}} + y_{n_{k_j}}) = \lim x_{n_{k_j}} + \lim y_{n_{k_j}} = x + y > 1 + 1 = 2.$$

Hence, we have found a subsequence of $(x_n + y_n)$ converging to something different from 2. This contradicts the assumption that $\lim(x_n + y_n) = 2$. □